

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

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October 2026

Micro- & Macroprudential Management · Country implementation of a CCyB increase

Master of Financial Technology · Frankfurt School of Finance & Management

The Netherlands raised its countercyclical capital buffer (CCyB) to 2% when the rulebook said 0% — refilling a buffer, not fighting a boom

Summary

Decision: CCyB 0% → 1% (ann. May 2022) → **2%** (ann. May 2023, binding 31 May 2024), together with lower buffers for systemically important banks (O-SII).

Puzzle: the Basel reference indicator, the credit-to-GDP gap (credit above its long-run trend), stood at **-33pp** and **-47pp** at the two decisions → the buffer guide implied **0%**.

Q1 · Environment

Recovered economy, then war, energy shock and **11.6%** inflation; ECB hikes from Jul-2022.

Credit cold; housing hot (+19% y/y), then falling.

Banks strong (core capital ratio, CET1, 17.7%) and more profitable as rates rose.

Q2 · Criteria & risks

DNB framework (2022): **2% = “standard risk environment”**, calibrated on past crisis losses (€12bn), not on credit growth.

Insures against shocks no indicator predicts; housing left to other tools.

Keeps the March-2020 promise to restore releasable capital.

Q3 · Implications

Cheap: ≤ about one fifth of big banks' CET1 headroom.

A swap: structural buffers cut in 2020 and 2024.

No sign of contraction in credit or its price. Complements ECB tightening; reciprocated abroad; **€6.7bn** releasable.

Verdict: **normalisation, not tightening.**

Trade-off: relies on discretion and credibility; release never tested.

When COVID hit, neighbours released their buffers — the Netherlands had nothing to release

CCyB release announcements, March–April 2020



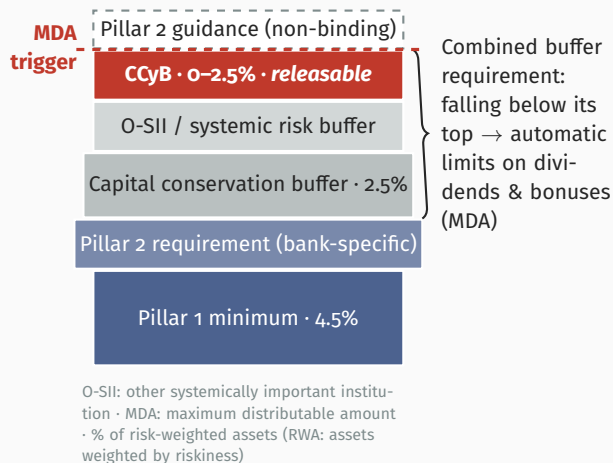
Netherlands: CCyB 0% — nothing to release

DNB improvised (17 Mar 2020):

- cut systemic buffers of ING, Rabobank, ABN AMRO: 3% → 2.5 / 2 / 1.5%
- postponed a planned floor on mortgage risk weights
- **€8bn** freed, up to **€200bn** of lending

... and made a promise

“Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%.”



Design

- 0–2.5% of risk-weighted assets, on *domestic* exposures
- Increases bind after 12 months; releases are immediate
- EU reciprocity: foreign banks lending in NL hold it too

Why “releasable” matters

- In COVID, banks close to the MDA trigger **cut lending** even when *allowed* to use buffers (stigma, payout limits)
- Only an explicitly **released** buffer is truly usable

DNB raised capital requirements in the middle of a storm, when its own rulebook said zero. Wrong moment?

The question

The storm (2022–23)

War in Ukraine · record energy prices
HICP inflation **11.6%** (2022)
ECB hikes from **21 Jul 2022**
House prices turning down
Basel gap **-33 / -47pp** → guide **0%**



The decision

CCyB **0% → 1% → 2%**
announced May 2022 and May 2023
binding May 2023 and May 2024

Raising capital in a downturn is the **procyclical mistake** — amplifying the cycle — that the CCyB was invented to avoid.

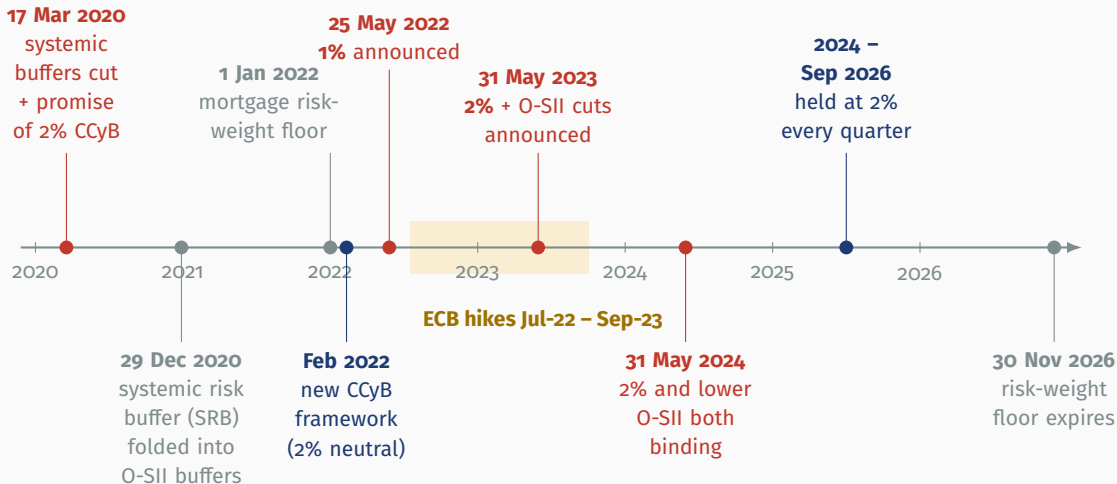
Tightening into a slowdown? Fighting a hidden boom? Or something else?

Act 1 · The setting (Q1 Environment)
timeline · macro
credit & housing · banks

Act 2 · The decision (Q2 Criteria & risks)
rule vs framework · DNB's reasons
risks & tools · verdict

Act 3 · The consequences (Q3 Implications)
banks · borrowers · monetary policy
borders · resilience · critiques

The path to 2% was announced in 2020 — and walked step by step as the ECB began to tighten



Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 17 Sep 2026; 24 Apr 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

The economy had fully recovered — then war, energy and inflation darkened the outlook

Q1 · Environment

2021Q3

GDP back above
pre-COVID level

11.6%

HICP inflation, 2022
(national CPI 10.0%)

+50bp

ECB's first hike
21 July 2022

record

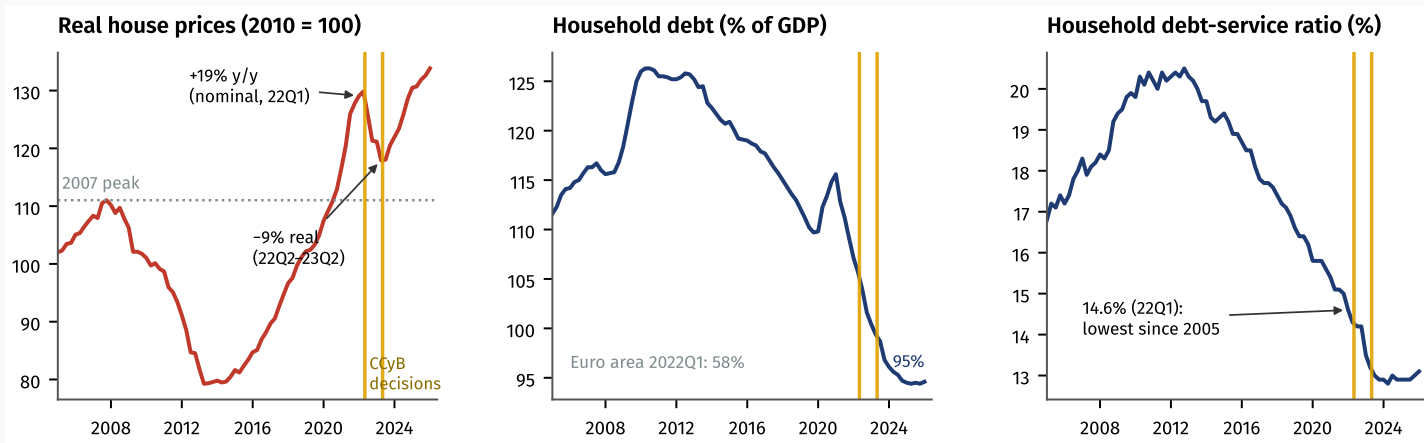
energy & commodity prices
after the invasion

DNB, Financial Stability Report, spring 2022

“After a strong recovery from the pandemic, the economic outlook has worsened due to the war in Ukraine and high inflation, and financial stability risks have increased.”

“Higher funding costs may put pressure on the debt sustainability of governments, businesses and households.”

By end-2021 the Dutch economy was almost 3% larger than at end-2019 — the recovery DNB's framework requires before building the buffer.



Housing: nominal prices +19% y/y (22Q1), real 16% above the 2007 peak. DNB itself: *“the overheated housing market calls for further measures.”*

Credit: household debt high (107% of GDP vs 58% in the euro area) but *falling*; debt service at its lowest since 2005.

Banks entered the storm well capitalised — and rising rates were lifting their profits

Q1 • Environment

17.7%

core capital (CET1) ratio
end-2021 (EU 15.7%, 21Q3)

16.3%

CET1 ratio, end-2022
“in line with EU average”

+6.7%

net interest income 2022
(67.7% of income)

11.5%

4 major banks' CET1 after a
severe stress test (min. 8%)

Q1 in one line — against raising now

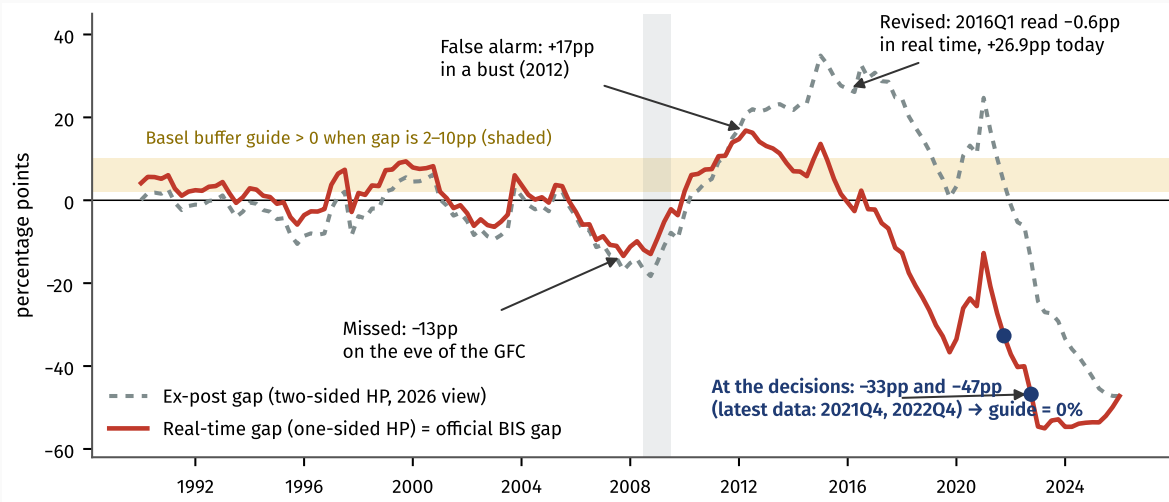
war and energy shock · 11.6% inflation
rapid ECB tightening · house prices turning

... and for raising now

economy recovered · credit cold
banks strong, profits rising with rates

DNB (FSR 2022): higher rates “*may ultimately have a positive impact on the profitability of financial institutions.*”

The Basel rule said 0% — but on Dutch data it had missed the last crisis and raised false alarms



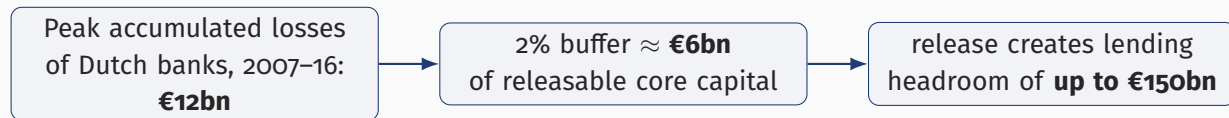
Credit-to-GDP gap = credit/GDP ratio minus its long-run trend (Hodrick-Prescott filter). **Rule:** gap < 2pp → 0%, gap > 10pp → 2.5%. Our one-sided filter ($\lambda = 400,000$) reproduces the official BIS gap within 0.00005pp (1971–2026).

Alternative trend filters are noisy around COVID (two-sided HP +4.2pp in 2021Q4) — one more reason not to trigger on any of them.

DNB's compass: 2% is the normal-times level, calibrated on past crisis losses — not on credit growth



built up 1pp a year → 2% after two years



- Criteria: a dashboard (macro, banks, households & firms, markets); *“no mechanical link ... guided discretion.”*
- 2% chosen *“also taking into account the impact of the buffer reduction in March 2020.”*
- Not alone: 17 jurisdictions run a *positive neutral* CCyB (kept filled in normal times); NL, Sweden, UK, Poland target 2%. ECB loss-based range: 1.1–1.8%.

DNB justified the steps by recovery, uncertainty and its 2020 promise — not by credit growth

Q2 · Criteria & risks

May 2022 · 0% → 1%

*“When the systemic risk buffers ... were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB.”*

“The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but ... the robust economic recovery ... provides grounds for a gradual build-up.”

Cyclical risks: “normal” to elevated.

May 2023 · 1% → 2%

*“Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk.”*

Cited: investors' rising risk appetite · real-estate prices falling · debt sustainability of firms and governments · banks “robust, partly due to rising interest rates”.

Both decisions: **a normal-to-elevated risk picture + strong banks + the promise.** Credit growth is never the reason.

The CCyB insures against shocks nobody can forecast; the known hot spots got their own tools

Q2 · Criteria & risks

Financial-stability risks named by DNB (FSR 2022/23)

Instrument

War, energy and supply shocks · rapid tightening of financial conditions and market corrections · debt sustainability of governments, firms, households · bank turmoil abroad (SVB, Credit Suisse, 2023)

CCyB — releasable for *any* shock

“Overheated” housing · high mortgage debt of first-time buyers · interest-only loans

borrower limits (loan-to-value, loan-to-income), tax reform, **mortgage risk-weight floor** (Jan 2022 – Nov 2026)

Size and concentration of the large banks (sector **400%** → **280%** of GDP)

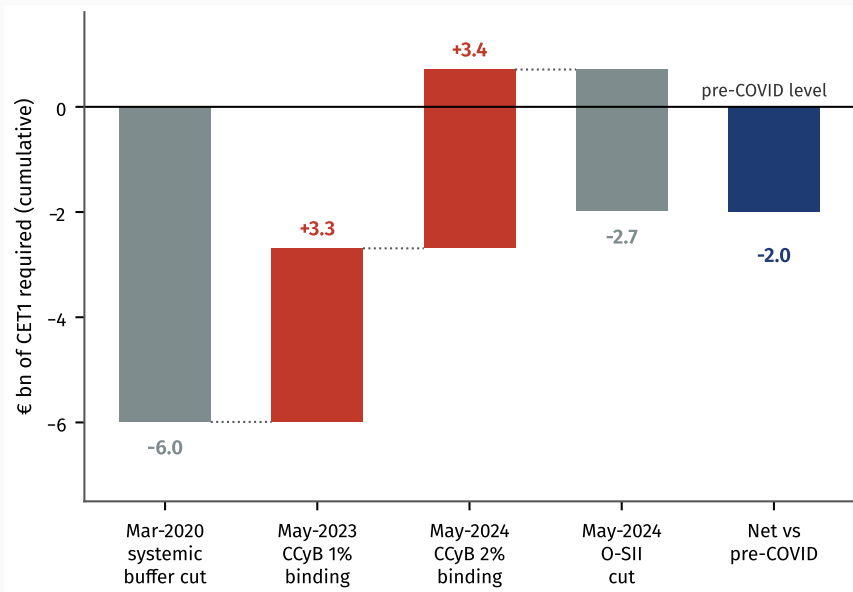
buffer for systemically important banks (**O-SII**, cut 2024)

Principle: one instrument per target (Tinbergen, 1952)

DNB on the 2020 recomposition: more releasable capital is “*a valuable addition given the sensitivity/volatility of the Dutch economy to external events.*” A releasable buffer is the only tool that works whatever the source of the shock.

If DNB were ...	normalising	fighting overheating	what happened	
Responding to credit indicators	no link	rises with them	gap, debt service, household debt ratio all <i>falling</i> ; CCyB <i>up</i>	✓
Path	+1pp a year, stop at 2%	follows risk	1% → 2% → held	✓
2023: house prices falling, rates jumping	raise anyway	pause	raised	✓
2024–26: house prices rising again (+10% y/y, Mar-26)	stay at 2%	go above 2% (framework phase 3)	held at 2%	✓
Other buffers	cut structural ones to compensate	keep them	cut in 2020 and 2024	✓

Five predictions, five matches: **normalisation**. The last row is the key to the whole story.



DNB's own words

"more or less capital-neutral for the three large banks" — 2020

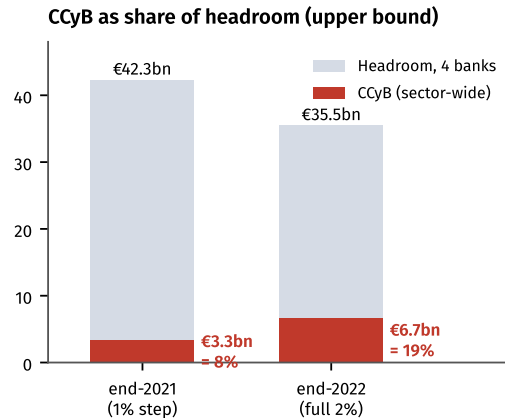
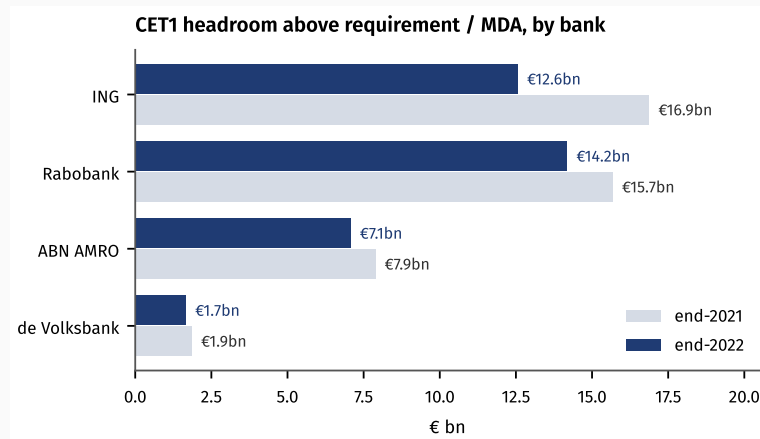
"a limited increase in the net capital requirements" — Knot, Jun 2023

"slightly increases the capital requirements for the banking sector as a whole" — FSR 2023

Not unusual: the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

Waterfall: approximate, own calculation. Buffer cuts valued at end-2022 total RWA of ING, Rabobank, ABN AMRO, de Volksbank (BNG excluded); CCyB amounts are DNB's sector-wide figures on Dutch exposures. Relative to 2021 the change was an increase.

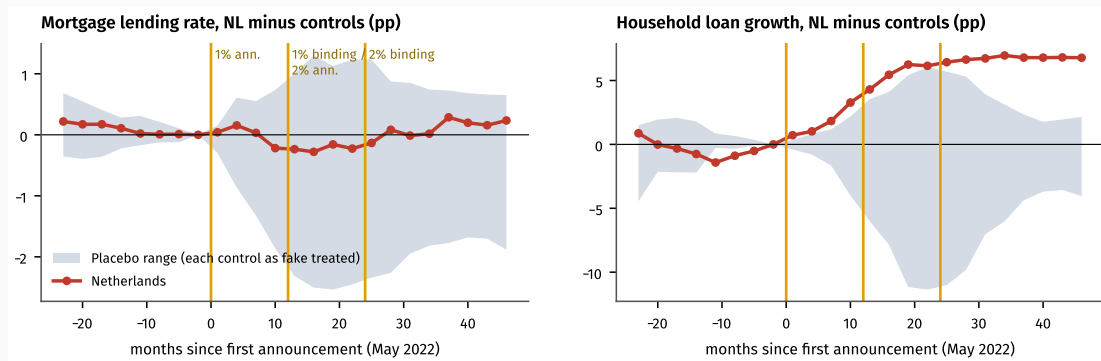
The twist: the storm made it the cheapest moment — higher rates gave banks the profits to build the buffer



Headroom = CET1 capital above the bank's requirement (MDA trigger). Net interest income **+6.7%** in 2022 → buffers built from *retained earnings*, not by cutting loans. Headroom shrank because of RWA growth and payouts, not the CCyB.

ECB: gradual build-up + high profitability “limit these economic costs” (MPB 24); activating buffers during the 2022–23 tightening “can mitigate implementation costs” (MPB 31).

Not the wrong moment — arguably the right one.



How to read: red = Netherlands minus countries that did not change their CCyB (AT, FI, IT, MT, LU); o = no difference; grey = differences that arise by chance (each control treated as if it had raised its CCyB).

- Dutch lending rates stayed **inside the placebo range** at all three CCyB dates
- Household credit grew **faster** than in peers (partly catch-up: NL lagged in 2019–22)

DNB (Sep 2026): “no signs of constraints in bank credit supply.”

One treated country, small control group, housing rebound: **no sign of contraction**, not a causal estimate.

Building the buffer while the ECB hiked did not double the squeeze — it let monetary policy focus on inflation

Q3 · Implications

The concern

Two tightenings at once: ECB rates *and* capital requirements → credit crunch?

What the data show

Dutch lending rates did not diverge from euro-area peers (event study); bank profits *rose* with rates (net interest income +6.7%).

One monetary policy for the euro area, but national macroprudential policy: the CCyB is the **Dutch-specific lever** — and the insurance against a rate shock gone wrong.

ECB Macroprudential Bulletin 31 (2025)

Early CCyB activation “helps monetary policy focus on its primary objective of price stability, thereby largely eliminating the potential for conflict between price stability and financial stability.”

DNB, FSR 2022

“A sudden tightening of financial conditions could also have a negative impact on financial stability.”

Through reciprocity every bank lending in the Netherlands holds the 2% — and Europe is moving the same way

Q3 · Implications

Across borders

- Foreign banks with Dutch exposures must also hold the 2% (mandatory reciprocity up to 2.5%)
- Limits leakage to foreign lenders — a known problem with national capital tools (Aiyar, Calomiris & Wieladek, 2014); *non-bank* leakage remains
- In April 2023, 5 banking-union countries had positive-neutral frameworks; NL had the **highest** rate (2%)

CCyB increases in the euro area

Country	Rate	Applies from
Germany	0.75%	Feb 2023
France	1.0%	Jan 2024
Ireland	1.5%	Jun 2024
Netherlands	2.0%	May 2024
Belgium	1.0%	Oct 2024
Portugal	0.75%	Jan 2026
Spain	1.0%	Oct 2026

€0 → €6.7bn

releasable-by-design capital
2020 → 2024

≈ half

of the €12bn peak losses
after the GFC

€150bn

lending headroom
if released

Asymmetric payoff (Lang & Menno, 2023)

Normal times: +1pp requirement → about **-0.1%** lending.

Bad times: 1pp release → up to about **+10%** lending.

Cheap to hold, powerful to release.

A growing role

The mortgage risk-weight floor expires on 30 Nov 2026.
DNB: this “*underpins the importance of ... the current countercyclical capital buffer (CCyB) of 2%.*”

Credibility	Without a mechanical rule, markets must trust DNB's judgement and communication — the classic rules-vs-discretion trade-off (Kydland & Prescott, 1977).
Calibration	Why 2%, not 1.5%? The ECB's loss-based range is 1.1–1.8%; NL chose the top of the banking union.
Untested	No release yet: will banks lend the freed capital — or hoard it, as with usable buffers in COVID?
Baseline	Relative to 2021 (not 2019) it was an increase: +€3.3bn before the O-SII offset.
Same profits, two claims	The ECB lists the Netherlands among countries adopting bank levies on excess profits in 2022–23 — partly claiming the earnings that fund the buffer.

A promise kept: the buffer was refilled in the storm — at the moment it was cheapest

Q1 · Environment

Recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.

Q2 · Criteria & risks

2% normal-times level, calibrated on past losses; insurance against unforeseeable shocks; housing to other tools; structural buffers cut in return.

Q3 · Implications

Cheap · largely a swap · no sign of contraction · complements monetary policy · reciprocated abroad · €6.7bn releasable · relies on discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it — even in a storm.

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake “treated” country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

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